

- Higher the status of species as pest more important the species is.
- Crop losses by the insect pests vary from 10-40 %

❖ **Different pesticides used:**

Insecticides	60%
Fungicides	19%
Herbicides	16%
Bio pesticides	2%
Others	3%

❖ **ORDERS OF INSECTS:**

Chilopoda	(Chilo - lip; poda - appendage) e.g. Centipedes
Diplopoda	(Diplo - two; poda - appendage) e.g. Millipede
Crustacea	(Crusta - shell) e.g. Prawn, crab, wood louse
Arachnida	(Arachne - spider) e.g. Scorpion, spider, tick, mite

❖ **CHARACTERS OF DIFFERENT ORDERS:**

Characters	Crustacea	Arachnida
Habit	Aquatic and few terrestrial	Terrestrial
Antenna	2 pair	No antenna
Visual organs	One pair Compound eyes	One pair-simple eyes
Locomotor organs	Minimum five pairs of legs	Four pairs
Respiration	Gill breathing	Book lungs (Scorpion) and tracheal(spiders)
Body fluid	Hemolymph	Hemolymph
Circulatory system	Heart with ostia	Heart with ostia

Characters	Chilopoda	Diplopoda	Hexapoda
Habit	Terrestrial	Terrestrial	Many terrestrial

			and very few aquatic
Antenna	One pair	One pair	One pair
Visual organs	One pair-simple eyes	One pair-simple eyes	Both simple eyes and compound eyes (one pair)
Locomotor organs	One pair per segment (First pair of legs modified as poison claws)	Two pair per segment (No poison claws)	
Respiration	Tracheal	Tracheal	Tracheal
Body fluid	Hemolymph	Hemolymph	Hemolymph
Circulatory system	Heart with ostia	Heart with ostia	Heart with ostia

❖ **Class Insecta has 2 sub-classes:**

1. Apterygota
2. Pterygota

❖ **Pterygota is further divided into**

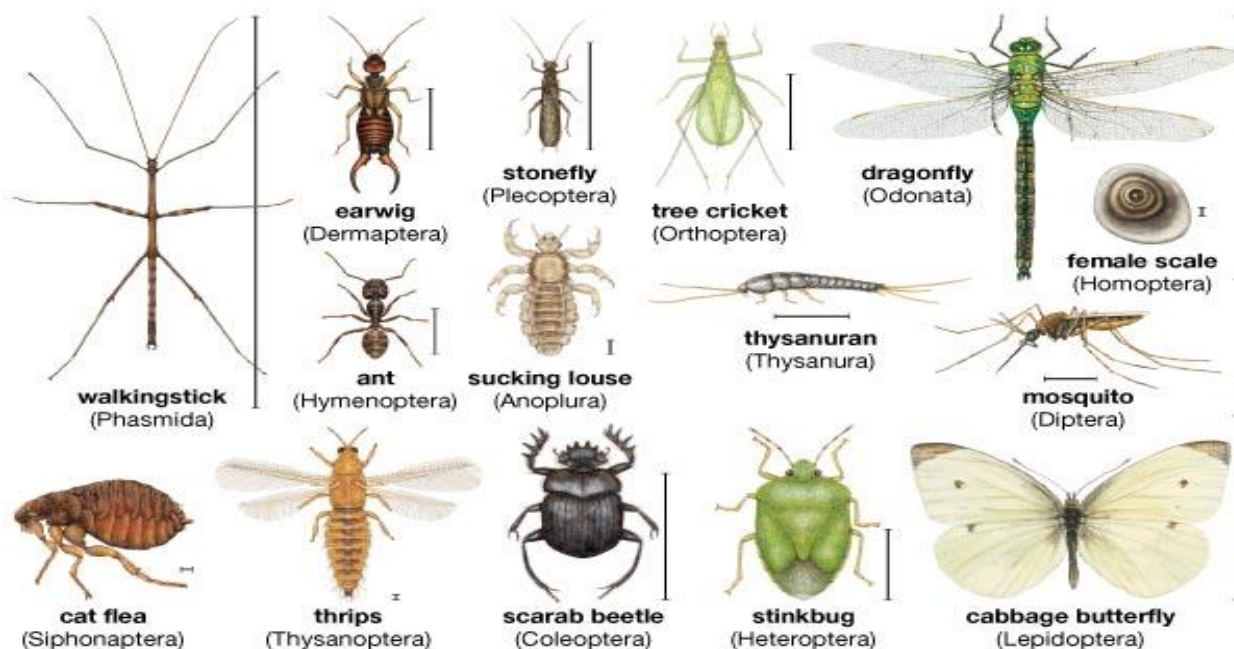
- Exopterygota
- Endopterygota

❖ **Apterygota:**

- Primarily wingless, metamorphosis absent, pregenital appendages present.
- It includes the orders:
 - ✓ **Protura (telsontails)**
 - ✓ **Diplura (japygids)**
 - ✓ **Collembola (springtails)**
 - ✓ **Thysanura (silverfish)**

❖ **Exopterygota-**

- Wings developed externally, metamorphosis incomplete, pupa absent, immature nymphal stage present.
- Incomplete metamorphosis- presence of 3 stages; eggs, nymph and adult.



❖ Endopterygota:

- Wings developed internally, metamorphosis complete, pupa present, immature stage is larva.
- **Complete metamorphosis**- presence of 4 stages; egg, larva, pupa and adult.

❖ Orders of Exopterygota with their examples:

Orders	Examples
Orthoptera	Locust, Grasshopper
Isoptera	Termite
Thysanoptera	Thrips
Hemiptera	1. Heteroptera: True Bugs 2. Homoptera: Aphid, Leaf Hopper

❖ Orders of Endopterygota with their examples:

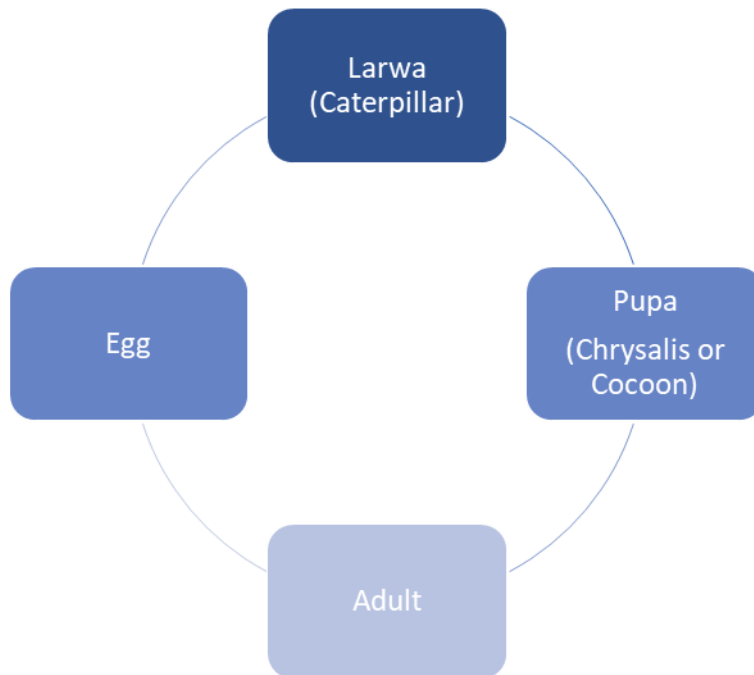
Orders	Examples
Coleoptera	Beetles and Weevils (This order is most damaging)
Lepidoptera	Butterflies and Moths
Diptera	Flies
Hymenoptera	Honeybee, Saw Flies, Ants, Wasps

❖ **Metamorphosis:** A profound change in form one stage to the next in the life history of an organism.

❖ **Types of metamorphosis:**

Ametabolous	No metamorphosis
Hemimetabolous	Incomplete metamorphosis
Holometabolous	Complete metamorphosis

- **Simple Metamorphosis/ Ametamorphosis:** Hatches from egg looking just like the adult grows in size by molting.
- **Incomplete Metamorphosis/ Hemi-metamorphosis:** Hatches from egg and becomes a nymph. Nymph does not have fully developed wings. Molts to become an adult.
- **Complete Metamorphosis/ Holo-metamorphosis:**



❖ **Types of Larvae:**

- **Nymph-** Larva of Orthoptera (grasshoppers/locusts) and Hemiptera (**bugs, white flies, aphids and jassids**).

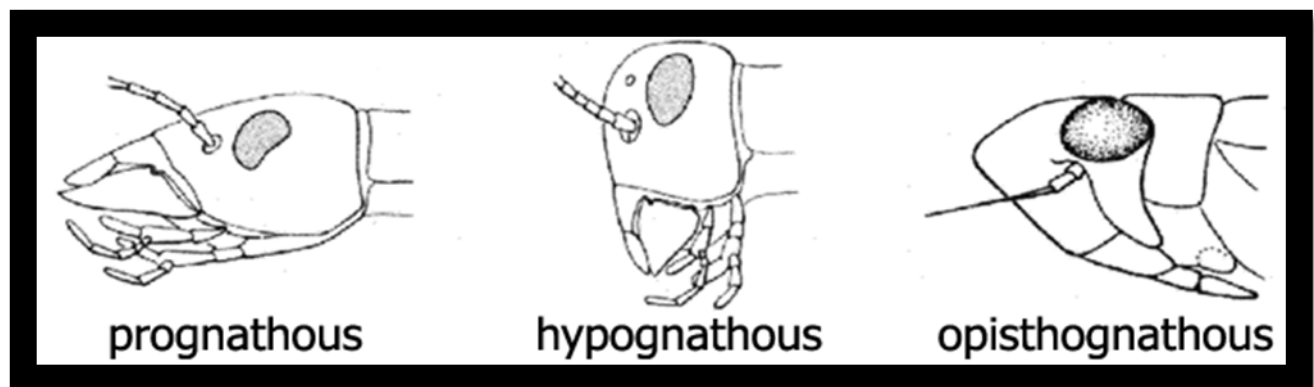
- **Caterpillar**- Larva of Lepidoptera (**butterflies/moths**), **Bollworm, borer** (except lesser grain borer).
- **Grub**- Larva of Coleoptera (**beetles/ weevils**) and Hymenoptera (**bees. Wasps, ants**).
- **Maggot**- Larva of Diptera (**true flies**)
- **Niads**- Larva of Odonata (**dragonfly/damselfly**)

❖ **The Insect Head:**

- The head of an insect is an almost completely sclerotized capsule formed by the fusion of several sclerites often distinctly seen in the **embryo**.

❖ **Based on the inclination of the long axis and the mouth parts, it is usual to distinguish three types of heads.**

- **Hypognathous** head with the mouth parts ventrally placed the long axis of the head being vertical. The median line of the head forms right angle with the median line of the body. **Eg. Grasshopper.**
- **Prognathous head**- Here the long axis is horizontal, and mouthparts placed anteriorly. **Eg. Beetles.**
- **Opisthognathous head**- mouthparts are posteroventral (placed between the anterior pair of legs) and head directed backwards. **E.g., Bugs.**



- ❖ **Antennae**: Antennae are mobile, segmented, paired appendages. Primitively, they appear to be eight segmented in nymphs and adults, but often there are numerous subdivisions, sometimes called antennomeres.